

MTE 204: Numerical Methods

3.0 Systems of Linear Equations

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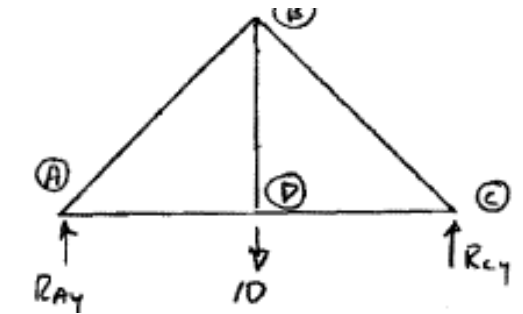
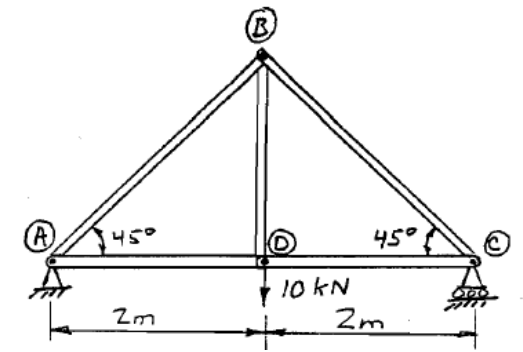
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3.0 Systems of Linear Equations [Roots]

Introduction

- Previously, we investigated methods for finding the root(s) of a single equation
 - Found the value x that satisfied a single equation $f(x)=0$
 - Methods: Graphical, Bisection, Fixed Point Iteration, Newton-Raphson, Secant methods
- Many engineering problems can be expressed as a series of linear equations.
 - Problems with 3 or fewer equations can be solved by hand.
 - Higher numbers of equations require the use of a computer.
- Methods for solving systems of linear equations:
 - Graphical
 - Direct methods
 - Cramer's Rule
 - Elimination of unknowns
 - Gaussian Elimination (w/ Pivoting and Scaling)
 - Indirect methods (iterative)
 - Gauss-Seidel (w/ Relaxation)



$$\Sigma F_y = 0$$

$$R_{Ay} + R_{Cy} = 10 \text{ kN}$$

$$\Sigma M_A = 0$$

$$0 R_{Ay} + 4 R_{Cy} = 20 \text{ kNm}$$

$$R_{Ay} = 5 \text{ kN}$$

$$R_{Cy} = 5 \text{ kN}$$

In matrix form

$$[A]_{2 \times 2} \{X\}_{2 \times 1} = \{B\}_{2 \times 1}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix} \begin{Bmatrix} R_{Ay} \\ R_{Cy} \end{Bmatrix} = \begin{Bmatrix} 10 \\ 20 \end{Bmatrix}$$

$\{ \} = \text{Column vector}$

3.0 Systems of Linear Equations [Roots]

- We seek values $x_1, x_2 \dots x_n$ that satisfy multiple linear equations.
- Linear equations
 - e.g. No powers (exponents) on x , no products of x
- We need the same number of equations as unknowns (e.g. $n \times n$ matrix for $[A]$)

General Form:

$$a_{11}x_1 + a_{12}x_2 + \dots a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots a_{2n}x_n = b_2$$

....

$$a_{n1}x_1 + a_{n2}x_2 + \dots a_{nn}x_n = b_n$$

$a_i, b_i = \text{constants}, x_i = \text{unknowns}$

$n = \text{\#equations, \# unknowns}$

In matrix form

$$[A]_{n \times n} \{X\}_{n \times 1} = \{B\}_{n \times 1}$$

$\{\} = \text{Column vector}$

3.1 Systems of Linear Equations [Graphical Method]

- Graphical Method – consider a set of 2 equations

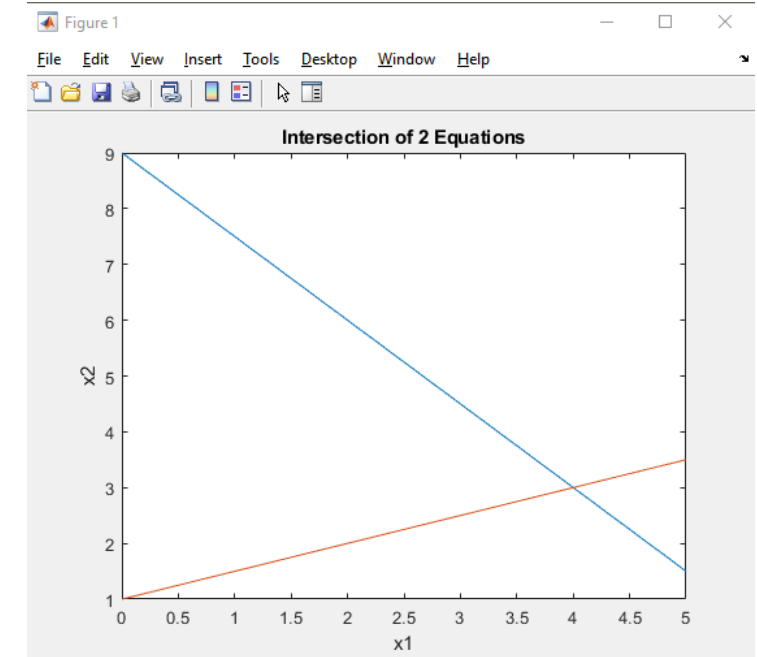
$$a_{11}x_1 + a_{12}x_2 = b_1 \quad \Rightarrow \quad 3x_1 + 2x_2 = 18$$

$$a_{21}x_1 + a_{22}x_2 = b_2 \quad \Rightarrow \quad -x_1 + 2x_2 = 2$$

Solve each equation for x_2 (isolate x_2), then plot x_2 versus x_1 and find the intersection point.

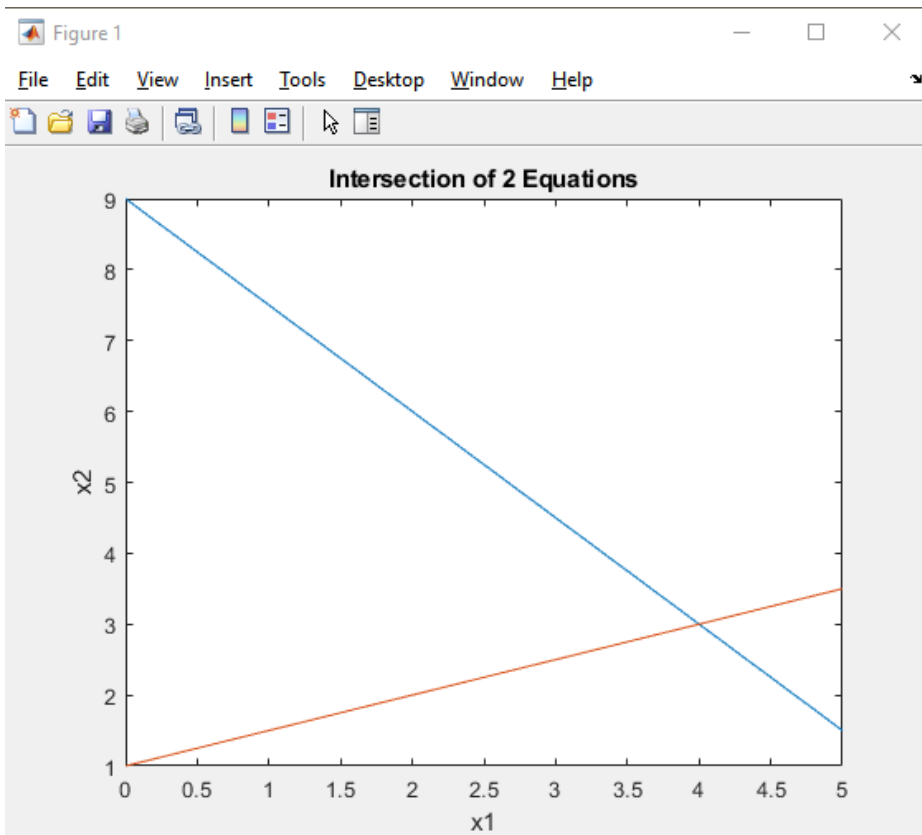
$$3x_1 + 2x_2 = 18 \quad \Rightarrow \quad x_2 = \frac{18 - 3x_1}{2} = 9 - \frac{3x_1}{2}$$

$$-x_1 + 2x_2 = 2 \quad \Rightarrow \quad x_2 = \frac{2 + x_1}{2} = 1 + \frac{x_1}{2}$$



3.1 Systems of Linear Equations [Graphical Method]

- Plotting the functions



Intersection

$$1 + \frac{x_1}{2} = 9 - \frac{3x_1}{2}$$

$$x_1 = 4$$

Substitute into
original equation

$$x_2 = 1 + \frac{x_1}{2}$$

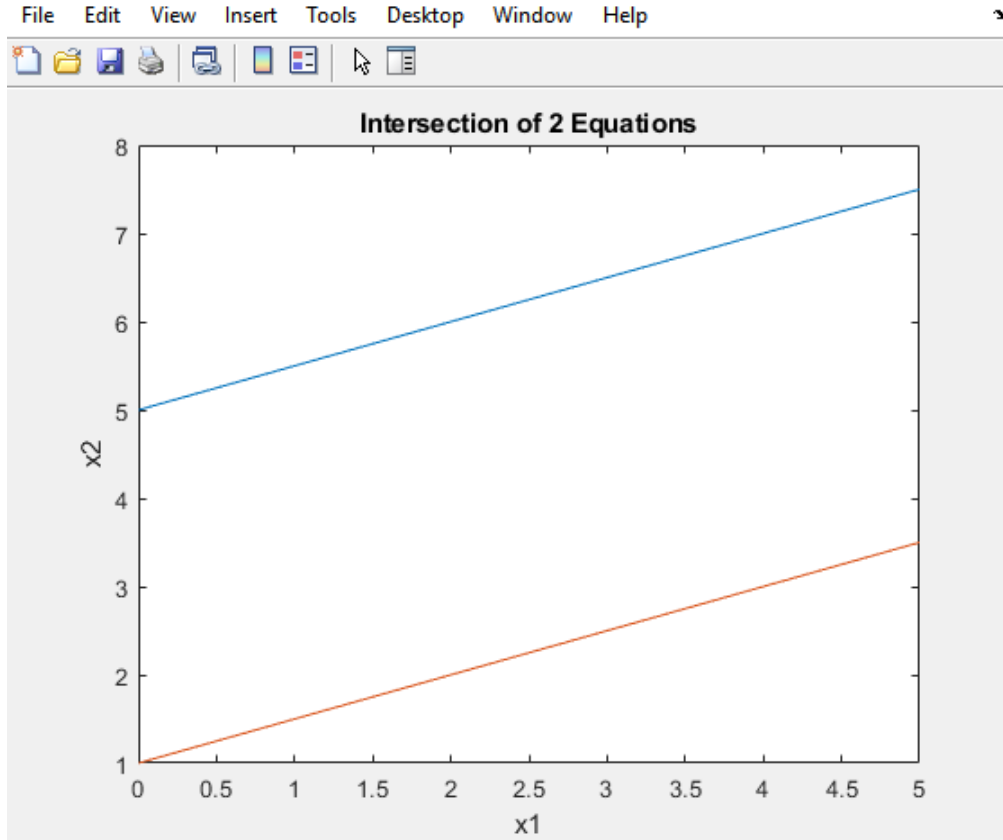
$$x_2 = 3$$

$$x_2 = 9 - \frac{3x_1}{2}$$

$$x_2 = 1 + \frac{x_1}{2}$$

3.1 Systems of Linear Equations [Graphical Method]

- What if the lines are parallel?



$$x_2 = 5 + \frac{x_1}{2}$$

$$x_2 = 1 + \frac{x_1}{2}$$

* Later we will see that lines with very similar slopes may also present challenges due to round-off.

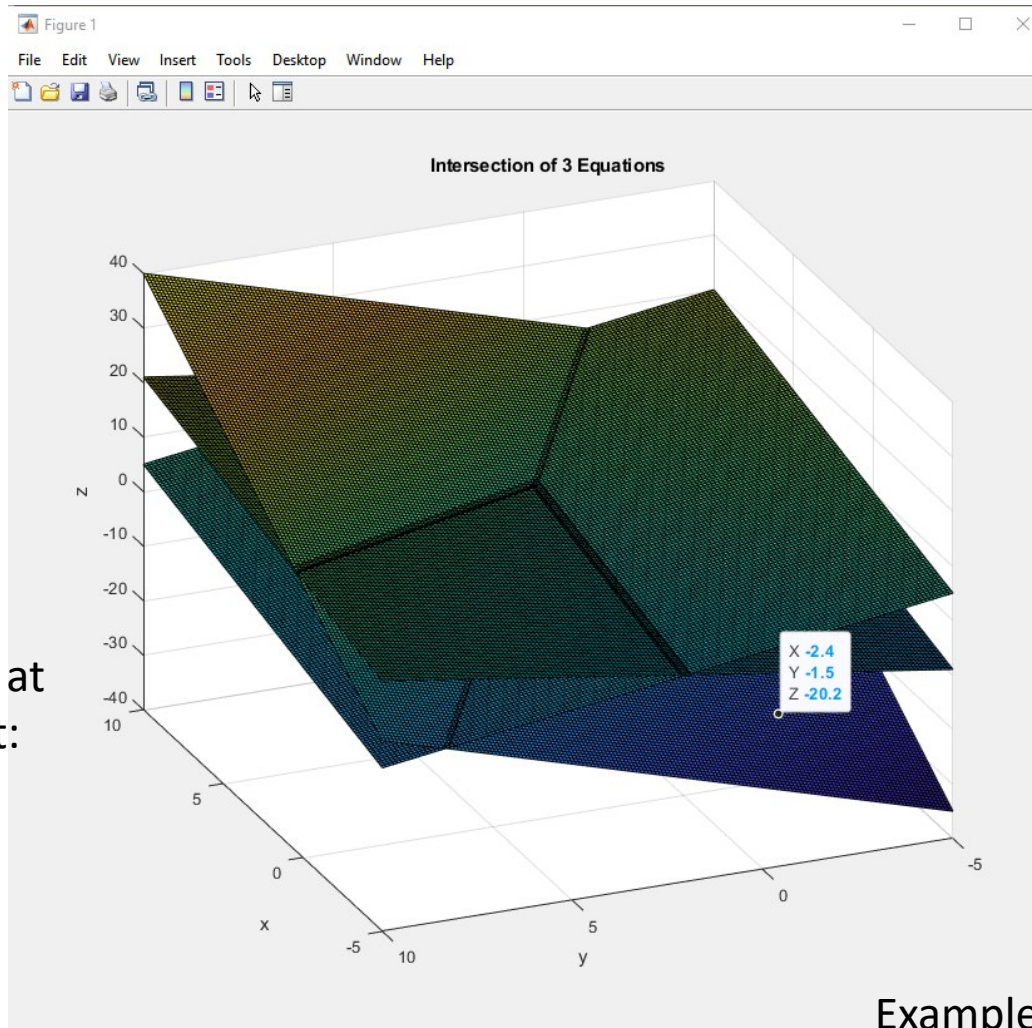
3.1 Systems of Linear Equations [Graphical Method]

- Consider 3 equations and 3 unknowns $z=f(x,y)$:

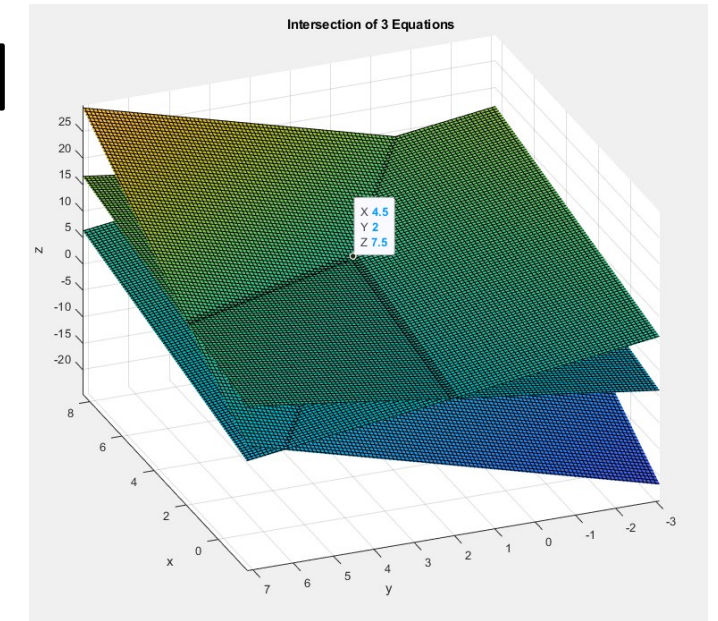
$$z=x-y+5$$

$$z=x+y+1$$

$$z=3*x+2*y-10$$



Example s4.m



Solution is the intersection of 3 planes at a unique point.

Matlab:

```
[x y] = meshgrid(-5:0.1:10)
z=x-y+5
surf(x, y, z)
hold on
z=x+y+1
surf(x, y, z)
hold on
z=3*x+2*y-10
surf(x, y, z)
hold on
title("Intersection of 3
Equations");
xlabel("x");
ylabel("y");
zlabel("z");
```

Clicking on the graph at the intersection point:

$x=4.5$

$y=2$

$z=7.5$

(approximate sol'n)

3.2 Systems of Linear Equations [Cramer's Rule]

- Cramer's rule is a formula for the solution of a set of linear equations. It can be used for a small number of equations (2 or 3), higher numbers becomes computationally expensive/complex to implement.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

We express the equations and unknowns as matrices.

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_2 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_2 = b_3$$

Let's rewrite this in matrix form:

$$[A] \{X\} = \{B\}$$

This rule states that each unknown in a system of linear algebraic equations may be expressed as a fraction of two determinants, with denominator D and with the numerator obtained from D by replacing the column of coefficients of the unknown in question by the constants $b_1, b_2, \dots, b_n$.

Essentially similar to solving by eliminating variables, but uses the determinants of the matrices.

3.2 Systems of Linear Equations [Cramer's Rule]

- The Determinant of [A] is a scalar (single) value.

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$D = |A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

3.2 Systems of Linear Equations [Cramer's Rule]

- Example (calculating the determinant):

$$0.3x_1 + 0.52x_2 + x_3 = -0.01$$

$$0.5x_1 + x_2 + 1.9x_3 = 0.67$$

$$0.1x_1 + 0.3x_2 + 0.5x_3 = -0.44$$

$$D = \begin{vmatrix} 0.3 & 0.52 & 1 \\ 0.5 & 1 & 1.9 \\ 0.1 & 0.3 & 0.5 \end{vmatrix}$$

3.2 Systems of Linear Equations [Cramer's Rule]



- Cramer's Rule

$$X_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{D}$$



$$= \frac{\begin{vmatrix} -0.01 & 0.52 & 1 \\ 0.67 & 1 & 1.9 \\ -0.44 & 0.3 & 0.5 \end{vmatrix}}{-0.0022}$$

$$= \frac{0.3278}{-0.0022}$$

$$X_1 = -14.9$$

$$0.3x_1 + 0.52x_2 + x_3 = -0.01$$

$$0.5x_1 + x_2 + 1.9x_3 = 0.67$$

$$0.1x_1 + 0.3x_2 + 0.5x_3 = -0.44$$

Replace the column #1 values of [A] with {B}
(values from the right hand side of the equations).

Calculate the determinant of the top, as in the
previous slide.

3.2 Systems of Linear Equations [Cramer's Rule]



- Cramer's Rule



$$X_2 = \frac{\begin{vmatrix} 0.3 & -0.01 & 1 \\ 0.5 & 0.67 & 1.9 \\ 0.1 & -0.44 & 0.5 \end{vmatrix}}{-0.0022}$$

$$X_2 = -29.5$$

For X_2 replace the column #2 values of [A] with {B}
(values from the right hand side of the equations).

Calculate the determinant of the top, as in the
previous slide.



$$X_3 = \frac{\begin{vmatrix} 0.3 & 0.52 & -0.01 \\ 0.5 & 1 & 0.67 \\ 0.1 & 0.3 & -0.44 \end{vmatrix}}{-0.0022}$$

$$X_3 = 19.8$$

For X_3 replace the column #3 values of [A] with {B}
(values from the right hand side of the equations).

Check:

$$0.5x_1 + x_2 + 1.9x_3 = 0.67$$

LHS

$$0.5(-14.9) + (-29.5) + 1.9(19.8) = 0.67 = \text{RHS} \quad \checkmark$$

3.3 Systems of Linear Equations [Elimination of Unknowns – manual calc]

- We can combine equations algebraically to eliminate one of the unknowns, and solve for the remaining unknown (2 equations).
- Example:

$$2x_1 + 3x_2 = 5 \quad (1)$$

$$5x_1 - 6x_2 = 1 \quad (2)$$

*** Always substitute into the original equations to verify the solution.

- This method works for 2 or 3 equations, but applying to larger numbers of equations is complicated and we generally require a computer to do the calculations.

3.4 Systems of Linear Equations [Gaussian Elimination]

- Solving systems of linear equations requires significant computations for more than 3×3 problems.
- Ideal problem for computer-based solution, where we may have thousands or millions of simple operations (e.g. addition, subtraction, multiplication, division).
- Gaussian Elimination is a direct method:
e.g. Gaussian Elimination, we solve for the 'exact' solution in one pass through the equations.
- Aside, later we will discuss Indirect, iterative methods
e.g. Gauss Seidel, we approximate the solution (initial guesses) and iterate to converge to a final solution within a tolerance level.

3.4 Systems of Linear Equations [Gaussian Elimination]

Step 1 – Forward elimination of unknowns

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

Clear column #1 below $a_{1,1}$

*** Called Naïve Gauss Elimination since we solve the equations 'as-received' and don't check for divide by zero.

*** Goal is to achieve an upper triangular matrix by forward elimination, and then solve by backwards substitution.

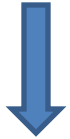
3.4 Systems of Linear Equations [Gaussian Elimination]

Step 2 - Apply the same procedure to Row 3

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b_3 \end{Bmatrix}$$

Prime (e.g. b') notation indicates a modified coefficient

Row #3 = Row #3 - (a₃₁/a₁₁) Row #1



$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & a'_{3,2} & a'_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \end{Bmatrix}$$

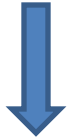
Goal is to zero the terms below the diagonal

3.4 Systems of Linear Equations [Gaussian Elimination]

Step 3 – Clear (zero) column #2

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & a'_{3,2} & a'_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \end{Bmatrix}$$

Row #3 = Row #3 – ($a'_{3,2}/a'_{2,2}$) Row #2



$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & 0 & a''_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \end{Bmatrix}$$

Forward elimination is complete when terms below the diagonal are zero.

Double prime (e.g. b'') notation indicates that the coefficient has been modified twice.

3.4 Systems of Linear Equations [Gaussian Elimination]

Backwards substitution

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & 0 & a''_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \end{Bmatrix}$$

Start at the bottom

$$a''_{33}x_3 = b''_3$$

Solve for $x_3 = b''_3 / a''_{33}$

Move up one row (2nd row):

$$a'_{22}x_2 + a'_{23}x_3 = b'_2$$

Substitute x_3 value and solve
for x_2

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

Substitute x_2 and x_3 values and
solve for x_1

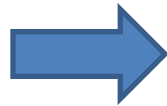
3.4 Systems of Linear Equations [Gaussian Elimination]

Example

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$



$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{Bmatrix}$$

Step 1: Forward Elimination (clear Column #1)

$$\text{Row \#2} = \text{Row \#2} - (a_{21}/a_{11}) \text{ Row \#1}$$

*** Carry all digits for
intermediate calculations

3.4 Systems of Linear Equations [Gaussian Elimination]

Example (Forward elimination)

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.29333 \\ 0.3 & -0.2 & 10 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 71.4 \end{Bmatrix}$$

Row #3 = Row #3 - (a_{31}/a_{11}) Row #1

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.29333 \\ 0 & -0.19 & 10.02 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.615 \end{Bmatrix}$$

3.4 Systems of Linear Equations [Gaussian Elimination]

Example (Forward elimination)

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.29333 \\ 0 & -0.19 & 10.02 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.615 \end{Bmatrix}$$

Clear Column #2

Row #3 = Row #3 - (a'_{32}/a'_{22}) Row #2

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.29333 \\ 0 & 0 & 10.012 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}$$

3.4 Systems of Linear Equations [Gaussian Elimination]

Example (Backwards Substitution)

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.29333 \\ 0 & 0 & 10.012 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}$$

*** Always substitute x's into the original equations to verify the solution.

3.4 Systems of Linear Equations [Gaussian Elimination]

This method is called Naïve Gaussian Elimination, because we are not checking for a divide by zero.

i.e. $\text{Row \#2} = \text{Row \#2} - (a_{21}/a_{11}) \text{ Row \#1}$

If the term on the diagonal (a_{11}) is zero, then we will have a divide by zero error.